

Discrete Mathematics (MATH 208)

Fall 2022

Friday, 11/18/2022

Homework 12: Chapters 30, 31

Score: / 10

Name (Print): _____

Chapter 30 - The Binomial Theorem and Pascal's Triangle

1 pts

1. Determine the coefficient of x^5y^2 in the expansion of $(2x - 5y)^7$.

2 pts

2. Give an algebraic proof that $\binom{n}{2}\binom{n-2}{m-2} = \binom{n}{m}\binom{m}{2}$.

2 pts

3. Give a combinatorial proof that $\binom{n}{2}\binom{n-2}{m-2} = \binom{n}{m}\binom{m}{2}$.

2 (bonus)

4. Determine the coefficient of $x^3y^2z^3$ in the expansion of $(x + 2y - 3z)^8$.

Chapter 31 - Inclusion-Exclusion Counting

1 pts

1. How many integers between 1 and 10000 (inclusive) are divisible by at least one of 9, 10, or 11?

2 pts

2. Suppose p, q are two different primes. How many integers between 1 and the product pq are relatively prime to pq (or, to rephrase things in a slightly different but equivalent way, how many integers between 1 and pq are divisible by neither of p and q)? (The correct answer will factor neatly.)

2 pts

3. Of the words of length ten using the alphabet $\Sigma = \{a, b, c\}$, how many either begin abc or end cba or have $ccccc$ as the middle six letters?